

REPOSITORY HARDENING REPORT

512 / CVS Evidence-Sidecar

May 1 Seal Documentation + Post-Seal Interface Design Pass

Report period	01 May 2026 — 10 May 2026
Prepared by	Jonathan M. Watson
Pass type	Dual-phase: May 1 Seal Documentation + Post-Seal Interface Design — Execution Boundary Interface specification, runtime interface lock, witness runtime interface alignment, canonical invariant mapping, VCP parallel-development acknowledgment
Repos affected	2 — 512 (JonathanMastersWatson/512), CVS Evidence-Sidecar (JonathanMastersWatson/Evidence-Sidecar)
Total files changed	8 new / updated files across two repositories
Canon docs	512_ARCHITECTURE v3.4 (unchanged) · 512_IMPLEMENTATION v3.3 (unchanged) · CVS_ARCHITECTURE v3.2 (unchanged) · CVS_IMPLEMENTATION v2.7 (unchanged)
Kernel	Unchanged — SHA-256: 7B08C024B77A24830C15E7952D6E54BED383AA960F4C74A71FF95CE51F4D80F5
512 Seal status	SEALED — May 1, 2026. Archive SHA-256: DDC508C498CF156B4A86C83EBBE54E3F643EBC786C8A9027C41D8CD9BA254E290
CVS Seal status	SEALED — May 1, 2026. Archive SHA-256: 70297B8808522B01B0A1E7822E550F4A21B742C16DA7D098218FB9FA3968040C

1. Executive Summary

This report covers two phases of work: (1) documentation of the May 1, 2026 seal events for both the 512 and CVS Evidence-Sidecar repositories, which were not accompanied by a hardening report at the time of sealing; and (2) the post-seal interface design work conducted between May 1 and May 10, 2026.

The May 1 seal closed the pre-hardening phase for both repositories. Both repos were sealed simultaneously at the same archive date, following a BUILDERS/ folder pass that consolidated all white papers, canon documents, and the enterprise integration guide into a single developer-facing reference directory. Both CANONICAL_COMMITMENT.md and PRE_HARDENING_NOTICE.md were updated to reflect sealed status. Pre-hardening banners were removed from all normative documents across both repos.

The post-seal period produced the most substantive interface design work to date. The Execution Boundary Interface (EBI) was formally specified — first in v1.0, then expanded significantly to v1.1 — establishing the complete contract between upstream callers and the 512 Commit Gate. The runtime interface was formally locked. The invariant mapping from runtime proxy identifiers (inv_1 through inv_7) to canonical invariants (K1 through K7) was completed and documented. The CVS Witness Runtime Interface was specified, establishing the boundary event input schema and aligning the witness layer's intake with the EBI output contract.

The kernel is unchanged. Both repos are sealed. No canon documents were version-bumped. The post-seal work adds interface-layer specification without modifying any normative architectural claim.

2. Seal Registry — Complete History

Date	Repository	Archive SHA-256	Status
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2026-03-24	512	DF27F6C5C8DDBBD5341FB15EA943D92B3388331B386C26F44A145673F6C8D218	Supersec
2026-03-24	CVS Evidence-Sidecar	0DA1ABA3A7257B636C0A364920C5CE643843257D217C0DBD68BA5DB64712BE17	Supersec
2026-05-01	512	DDC508C498CF156B4A86C83EBBE54E3F643EBC786C8A9027C41D8CD9BA254E290	CURRENT SEAL
2026-05-01	CVS Evidence-Sidecar	70297B8808522B01B0A1E7822E550F4A21B742C16DA7D098218FB9FA3968040C	CURRENT SEAL

The kernel hash is independent of all archive hashes and remains constant across all seal events:

SHA-256 (512-kernel-padded.txt):
7B08C024B77A24830C15E7952D6E54BED383AA960F4C74A71FF95CE51F4D80F5
XRPL Genesis Anchor TX: 378536A3CB75DECF90B6AE57F75292BDFF716285B01946870CAC158F8152D100

3. May 1 Seal — Changes Included at Time of Sealing

The following changes were committed and included in both sealed archives on May 1, 2026. This section provides the documentation record that was not issued at the time of sealing.

3.1 512 Repository — May 1 Seal Contents

File / Item	What It Does
CANONICAL_COMMITMENT.md (new)	Permanent priority record. Documents the 512 genesis commit date, kernel hash, XRPL anchor, and Apache 2.0 license declaration as the authoritative public record of origination. Append-only by design.
AARM_AND_512.md (new)	Architectural positioning document establishing 512's relationship to AARM (AI Assurance and Risk Management) and CSA (Cloud Security Alliance) frameworks. Defines 512 as enforcement infrastructure, not a framework — and the structural difference between the two.
PRE_HARDENING_NOTICE.md (updated)	Updated from active hardening marker to sealed status record. Records the hardening phase timeline: genesis (Dec 17, 2025), hardening open (Mar 2026), first seal (Mar 24), BUILDERS/ added (May 2026), hardening closed (May 2026).
LIVING_DOCUMENTS.md (updated)	Updated to reflect sealed release. Canon documents in BUILDERS/ identified as living documents subject to version bumps; frozen documents enumerated.
FAILURE_MODES.md (updated)	Pre-hardening banner removed. Document now carries authoritative status without qualification.
ANTI_DRIFT.md (updated)	Pre-hardening banner removed.
512-ops/COMMIT_BOUNDARY_REFERENCE.md (updated)	Pre-hardening banner removed.

BUILDERS/ (new folder — 8 documents)	Single developer-facing reference directory. Contains: README.md, 512_ARCHITECTURE_v3.4.md, 512_IMPLEMENTATION_v3.3.md, AARM_AND_512.md, 512_CVS_ENTERPRISE_v1_0.md, EXECUTION_BOUNDARY_PRINCIPLE.md, UNAUTHORIZED_BY_DESIGN.md, UNINSURABLE_BY_DESIGN.md. CVS canon docs (v3.2, v2.7) also co-located here.
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3.2 CVS Evidence-Sidecar Repository — May 1 Seal Contents

File / Item	What It Does
CANONICAL_COMMITMENT.md (new)	Permanent priority record for CVS. Documents genesis commit (December 17, 2025, commit 32cbd9b), VCP parallel development acknowledgment (VCP v1.0 published January 13, 2026 — 27 days after CVS genesis), and Apache 2.0 declaration.
VCP_AND_CVS.md (new)	Parallel development acknowledgment document. States that VeritasChain Protocol (VCP) and CVS independently arrived at the same sidecar witness architecture within 27 days. Documents five properties where CVS and VCP differ structurally: (1) external time anchoring — CVS uses XRP Ledger consensus vs VCP's PTPv2 network time; (2) fail-open as non-negotiable conformance requirement vs design intent; (3) three-plane structural separation; (4) normative linkage to 512 at the constraint gate layer; (5) selective disclosure architecture.
PRE_HARDENING_NOTICE.md (updated)	Updated to sealed status. Hardening phase timeline recorded: genesis Dec 17 2025, hardening open Mar 2026, canonical spec sealed Mar 24, BUILDERS/ added May 2026, hardening closed May 2026.
LIVING_DOCUMENTS.md (updated)	Sealed release declared for May 2026. Canon documents (CVS_ARCHITECTURE, CVS_IMPLEMENTATION, ANTI_DRIFT, CONFORMANCE) identified as living documents. Frozen documents enumerated: CANON_HASHES.md, CANONICAL_COMMITMENT.md, LEGAL_NOTICE.md, LICENSE, NOTICE, UPSTREAM.md, PRIMITIVE_BOUNDARY.md.
README.md, ANTI_DRIFT.md, CONFORMANCE.md, 00_INTENT/NON_GOALS.md (updated)	Pre-hardening banners removed from all four normative documents.
CANON_HASHES.md (updated)	Updated with May 1 seal hash for the CVS repo archive.
BUILDERS/ (new folder — 6 documents)	Developer-facing reference directory. Contains: README.md, CVS_ARCHITECTURE_v3.2.md, CVS_IMPLEMENTATION_v2.7.md, VCP_AND_CVS.md, 512_CVS_ENTERPRISE_v1_0.md, UNINSURABLE_BY_DESIGN.md.

4. Post-Seal Work — May 1 to May 10, 2026

The following work was conducted after the May 1 seal. These changes do not modify any sealed archive; they represent active development on the live repositories following the close of the pre-hardening phase.

4.1 512 Repository — Post-Seal Changes

File	Changes and Significance
docs/512_EBI_DESIGN_v1_0.md (new)	Initial Execution Boundary Interface design specification. Established boundary position, frozen interface contract (request shape, required fields, output tokens), invariant mapping (simplified), caller and gate obligations, CVS relationship, failure modes, and disambiguation table. This document formalized the EBI as a named, versioned specification — the first time the interface contract was expressed in a standalone document separate from the architecture and implementation canon.
docs/512_EBI_DESIGN_v1_1.md (new)	Major expansion of the EBI specification. Nine sections added, three sections expanded. Central deliverable: the complete invariant mapping table (§4) translating runtime proxy identifiers inv_1 through inv_7 to canonical invariants K1 through K7. This mapping had never existed as a single authoritative document — developers previously had no structured bridge between the reference runtime's field checks and the canonical invariant definitions in the architecture document. See §4.2 of this report for the full mapping.
runtime/INTERFACE_LOCK.md (updated)	Runtime interface formally declared frozen. Input contract (7 required fields: intent.action, intent.target, context.identity, context.consent, context.timestamp, context.system_state, constraints.max_amount) and output contract (ALLOW / DENY inv_N / DENY invalid_request / DENY evaluation_error) declared immutable. Breaking change definition established. Future Work item 'Replace placeholder invariant mappings with canonical 512 mappings' marked COMPLETE, referencing INVARIANTS.md and EBI v1.1 §4.
runtime/INVARIANTS.md (updated)	Cross-references added to docs/512_EBI_DESIGN_v1_1.md §4 as the authoritative proxy-to-canonical mapping document. Each invariant entry now carries: (1) the runtime check, (2) the canonical invariant it proxies, and (3) a note on what full production enforcement requires beyond the proxy. K6 and K7 notes updated to state explicitly that these are gate-behaviour invariants, not per-request field checks.

4.2 EBI v1.1 — Invariant Mapping (Central Deliverable)

The invariant mapping table in EBI v1.1 §4 is the most significant single artifact produced post-seal. It is reproduced here in full for the hardening record.

Runtime ID	Runtime check	Canonical invariant	What the proxy checks	What full enforcement requires
inv_1	context.identity is not None	K1 — No force or fraud against any human	Identity presence — necessary for attributing coercive action	Action classification (coercive/deceptive flag), transfer amount vs authorised limit, affected party type
inv_2	context.consent is True	K2 — Voluntary and explicit consent	Consent boolean — placeholder for documented explicit consent	Cryptographic consent token, consent type must be explicit (not implied), consent expiry not exceeded at evaluation time
inv_3	intent.action is not None	K3 — Consent may be withdrawn; exit must be possible	Declared action presence —	Epoch check: token epoch must match current consent registry

inv_4	intent.amount ≤ constraints.max_amount	K4 — Contracts must be explicit, readable, equally enforceable	precondition for evaluation Amount within declared limit — enforces one boundary of a declared term	epoch; withdrawal must have propagated Contract machine-readable AND human-readable, both parties acknowledged, no enforcement asymmetry
inv_5	context.timestamp is not None	K5 — No rules may be hidden or unilaterally changed	Timestamp presence — enables evidence sequencing	Active spec hash equals disclosed spec hash equals canonical hash (7B08C024...); disclosure acknowledged by all parties
inv_6	intent.target is not None	K6 — On failure, systems must fail open and disclose rules	Known target — proxy only; K6 is a gate-behaviour invariant, not a per-request field check	The fail-open handler IS the K6 implementation. There is no K6 field in the Proposal Object.
inv_7	context.system_state == 'healthy'	K7 — The specification is immutable; adherence is binary	System health as execution precondition — proxy only; K7 is a gate-startup invariant	SHA-256 hash verification at gate startup: sha256(spec) must equal 7B08C024... A gate that starts with hash mismatch refuses to start.

Amber rows (K6, K7) flag the two invariants that are structurally different from K1–K5. K1–K5 each enforce a property of the proposal. K6 and K7 enforce properties of the gate itself. Neither contributes a field to the EBI request.

4.3 CVS Evidence-Sidecar Repository — Post-Seal Changes

File	Changes and Significance
witness-runtime/WITNESS_INTERFACE.md (new/updated)	Witness Runtime Interface specification. Defines the boundary event input schema received by the CVS witness runtime from the 512 gate — aligning the witness layer's intake contract with the EBI v1.1 output contract. Key additions: invariant_result field with three values (ALL_PASS on ALLOW; inv_N on DENY indicating which invariant failed; evaluation_error on gate internal failure). Runtime responsibilities enumerated (8 MUSTs). Runtime non-responsibilities enumerated (8 MUST NOTs). Structural boundary restated: witness sits beside the gate, not inside it.

5. EBI v1.1 — Sections Added

Section	Content and Significance
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§2	Authorisation Precedes the Gate — Establishes that authorisation (identity verification, permission check) is an upstream concern. By the time a request arrives at the EBI, authorisation is established and carried as a cryptographic proof. The gate evaluates constraints, not identity. context.identity is a field the gate receives — it does not validate that the identity is genuine.
§4	Invariant Mapping — Central deliverable. Complete proxy-to-canonical table (inv_1–7 to K1–K7). Proxy check stated. What full production enforcement requires stated. K6 and K7 explicitly identified as gate-behaviour invariants with no corresponding Proposal Object fields. Cross-references to 512_ARCHITECTURE v3.4 §4 and 512_IMPLEMENTATION v3.3 §3 for each invariant.
§7	Evaluation Sequence — ASCII flow diagram showing early-exit evaluation order from required-field validation through inv_1 to inv_7 to ALLOW. Early exit is by design: first invariant to fail terminates evaluation. DENY inv_2 does not reveal state of inv_3–7.
§8	Fail-Open Mechanics — Clarifies that GAP is a witness-layer concept, not a gate output token. Three-step fail-open sequence documented. evaluation_error as production vs reference runtime distinction: reference runtime denies on error (fail-safe); production deployments implement fail-open handler (K6 compliance). An unhandled exception producing DENY in production is a K6 violation.
§9	Non-Conformant Integration Patterns — Five patterns A through E cross-referenced from 512_IMPLEMENTATION v3.3 §1.6. API handoff (A), queue handoff (B), broker handoff (C), pre-check positioning (D), parallel path (E). All create paths to irreversible state change that bypass gate evaluation.
§10	Common Developer Errors — Seven-row table covering: treating DENY as advisory, caching ALLOW, building bypass for errors, treating evaluation_error as retry, reading inv_N as the full invariant, implementing K6/K7 as request fields, positioning the gate upstream (Pattern D).
§11	Integration Checklist — Three-phase checklist: Pre-Integration (5 items), Per-Call (5 items), Post-Denial (5 items). First integration reference document in the entire 512 corpus structured as an actionable developer checklist.
§12	What the EBI Is Not — Disambiguation table with five rows covering: not a policy engine, not an advisory layer, not an audit log, not a pre-check, not runtime-configurable.
§13	Normative References — Complete reference table: 512_ARCHITECTURE v3.4, 512_IMPLEMENTATION v3.3, 512-ops/INTEGRATION_STEPS.md, 512-ops/CONSTRAINT_DEFINITION_LAYER.md, 512-ops/REFERENCE_FLOW.md, INTERFACE_LOCK.md, evaluator.py, CVS_ARCHITECTURE.

6. Interface Lock — Frozen Runtime Contract

The runtime interface lock, effective as of this period, establishes the following frozen contract surface. Any modification to these elements constitutes a breaking change requiring explicit versioning and a new interface surface.

6.1 Input Contract (Frozen)

Field	Type	Failure if absent or null
intent.action	string	DENY invalid_request
intent.target	string	DENY invalid_request
context.identity	string	DENY invalid_request

context.consent	boolean	DENY invalid_request
context.timestamp	string (ISO 8601)	DENY invalid_request
context.system_state	string	DENY invalid_request
constraints.max_amount	number	DENY invalid_request

6.2 Output Contract (Frozen)

Token	Meaning
ALLOW	All seven invariants evaluated TRUE. The commit path opens.
DENY inv_<N>	Invariant N evaluated FALSE. N is 1–7. First failure terminates evaluation. Commit path remains closed.
DENY invalid_request	Request was malformed before evaluation began. A required field is absent, null, or wrong type. No invariants were evaluated.
DENY evaluation_error	Gate encountered an unexpected internal error. Treat as DENY. Production deployments MUST implement fail-open handler rather than producing this token — an evaluation_error in production that results in DENY is a K6 violation.

7. CVS Witness Runtime Interface — Boundary Event Schema

The Witness Runtime Interface specification establishes the minimum boundary event the CVS witness must receive from 512. This schema is the handoff point between the gate's output contract (EBI) and the witness layer's intake contract (CVS).

Field	Type	Notes
proposal_id	string	Unique identifier for the proposal evaluated
decision_id	string	Unique identifier for this evaluation decision
runtime_id	string	Identifier of the gate runtime instance
decision	ALLOW DENY	Gate decision — two values only
invariant_result	ALL_PASS inv_N evaluation_error	ALL_PASS on ALLOW; inv_N on DENY (N = 1–7); evaluation_error on gate internal failure
invariant_id	string null	null on ALLOW; the failing invariant ID on DENY inv_N
proposal_hash	string	Hash of the full Proposal Object
decision_hash	string	Hash of the decision record
specification_hash	string	Must equal 7B08C024... — the canonical kernel hash
boundary_monotonic_timestamp	string	Monotonic timestamp at the execution boundary

8. VCP and CVS — Parallel Development Acknowledgment Summary

VCP_AND_CVS.md documents five structural properties where CVS and VCP diverge, beyond their shared sidecar witness architecture. This table summarises the differentiation for the hardening record.

Property	CVS	VCP
External time anchoring	XRP Ledger consensus — no single operator controls; impossible to backdate after finality	PTPv2 / NTP — high precision but operator-administered infrastructure
Fail-open	Non-negotiable conformance requirement across full failure mode enumeration	Design intent — conformance requirement status not formally stated in VCP specification
Structural separation	Three-plane architecture: Capture, Access, Interpretation — hard separation by design	Not stated in equivalent structural terms in published VCP specification
Normative linkage	Normatively linked to 512 at the constraint gate layer — CVS is the witness layer for a defined gate primitive	General-purpose sidecar — not linked to a specific constraint gate primitive
Selective disclosure	Formal selective disclosure architecture — proof of existence without revealing content	Not described as a formal architecture component in published specification

CVS genesis: December 17, 2025 (commit 32cbd9b). VCP v1.0 published: January 13, 2026. Priority gap: 27 days. Both establish the same base architecture independently — confirming it is a discovered natural solution, not an individual invention. Honest acknowledgment of parallel development strengthens the open commons claim.

9. Kernel Integrity Confirmation

Kernel text	7 statements — unchanged across all hardening passes
Canonical artifact	512-core/KERNEL/512-kernel-padded.txt
SHA-256	7B08C024B77A24830C15E7952D6E54BED383AA960F4C74A71FF95CE51F4D80F5
XRPL genesis anchor	378536A3CB75DECF90B6AE57F75292BDFF716285B01946870CAC158F8152D100
Invariant count	7 — confirmed in kernel text, INVARIANTS.md, CANONICAL_IR.json, and EBI v1.1 §4
CANONICAL_IR rules	7 (I1–I7) — corrected in April 23 pass from 9; unchanged this period
EBI runtime proxies	7 (inv_1–inv_7) — all mapped to canonical invariants in EBI v1.1 §4
Canon doc versions	512_ARCHITECTURE v3.4 · 512_IMPLEMENTATION v3.3 · CVS_ARCHITECTURE v3.2 · CVS_IMPLEMENTATION v2.7 — all unchanged this period

10. What Remains Open

Item	Notes
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CVS archive hash for post-seal state	LIVING_DOCUMENTS.md in CVS states 'a new archive hash will be committed to record the sealed state.' This hash has not yet been committed. The witness runtime interface additions (post-May 1) are not yet covered by any archive hash.
512 archive hash for post-seal state	The EBI v1.0, EBI v1.1, and INTERFACE_LOCK.md updates post-date the May 1 seal. A new archive hash should be committed for the current state if a third seal point is desired.
Production fail-open handler specification	EBI v1.1 §8.3 notes the reference runtime denies on evaluation exception (fail-safe) while production deployments MUST implement the fail-open handler per 512_IMPLEMENTATION v3.3 §3.6. No standalone fail-open handler specification document exists. Explicitly in-scope for future work.
Full K2 consent token specification	EBI v1.1 §4 notes that full K2 enforcement requires a cryptographic consent token with explicit type and expiry check. The reference runtime uses a boolean proxy. No standalone consent token schema has been published.
CVS custody architecture patterns	Explicitly out-of-scope frontier — not an omission.
Async saga correlation mechanics	Explicitly out-of-scope frontier — not an omission.
Multi-party constraint negotiation	Explicitly out-of-scope frontier — not an omission.
CA/ROLES/Career Transitions Into Constraint Architecture.md	Carried from April 23 report — still references '512-Compliant Consultant' (deprecated role name). Needs update to '512 Integration Specialist'.

11. Closing Statement

Both repositories are sealed. The kernel is unchanged. The pre-hardening phase is closed across both repos.

The post-seal work has done what the architecture required next: it connected the gate to the developer. The EBI v1.1 invariant mapping is the document that should have existed since the first time anyone tried to build against the reference runtime — the bridge from `inv_2` to K2, from a boolean field check to a cryptographic consent token with explicit type and expiry. It now exists. The interface is locked. The fail-open mechanics are documented. The non-conformant patterns are named and diagrammed.

The CVS witness layer's intake contract now mirrors the gate's output contract. The `invariant_result` field (`ALL_PASS`, `inv_N`, `evaluation_error`) closes the semantic gap between what the gate returns and what the witness records. The VCP parallel development acknowledgment strengthens both the prior art position and the open commons claim.

The architecture is harder to misintegrate than it was on May 1. The interface is stable. The truth it enforces is unchanged.

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10 May 2026